SE4031 – Games Development
2D Interactive Game Development

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Star Wings: Flyer

Star Wings is an immersive, infinite 2D side-scrolling action game. Players take command of a space explorer navigating a high-stakes, hazardous environment. The primary objective is survival; players must skillfully move vertically to dodge obstacles, engage and shoot down hostile enemies, and strategically collect health packs. To keep the gameplay engaging, the difficulty scales dynamically, featuring increasing player speeds and shifting spawn rates that provide a progressively intense challenge.

1. Core Features Implementation

The game's mechanics are built using modular C# scripts to meet all assignment requirements:

- **Health System:** Managed via PlayerController.cs, the health system features a dynamic UI slider with a color gradient that shifts from Green to Red as health depletes. Damage is registered upon collision with both environmental obstacles and enemies.

- **Scoring System:** The score is displayed using TextMeshPro on the UI. Points are awarded when the player successfully navigates through specific zones or destroys enemy units.

- **Health Packs:** These are randomly spawned collectibles that restore the player's energy levels upon contact, which is essential for surviving longer runs.

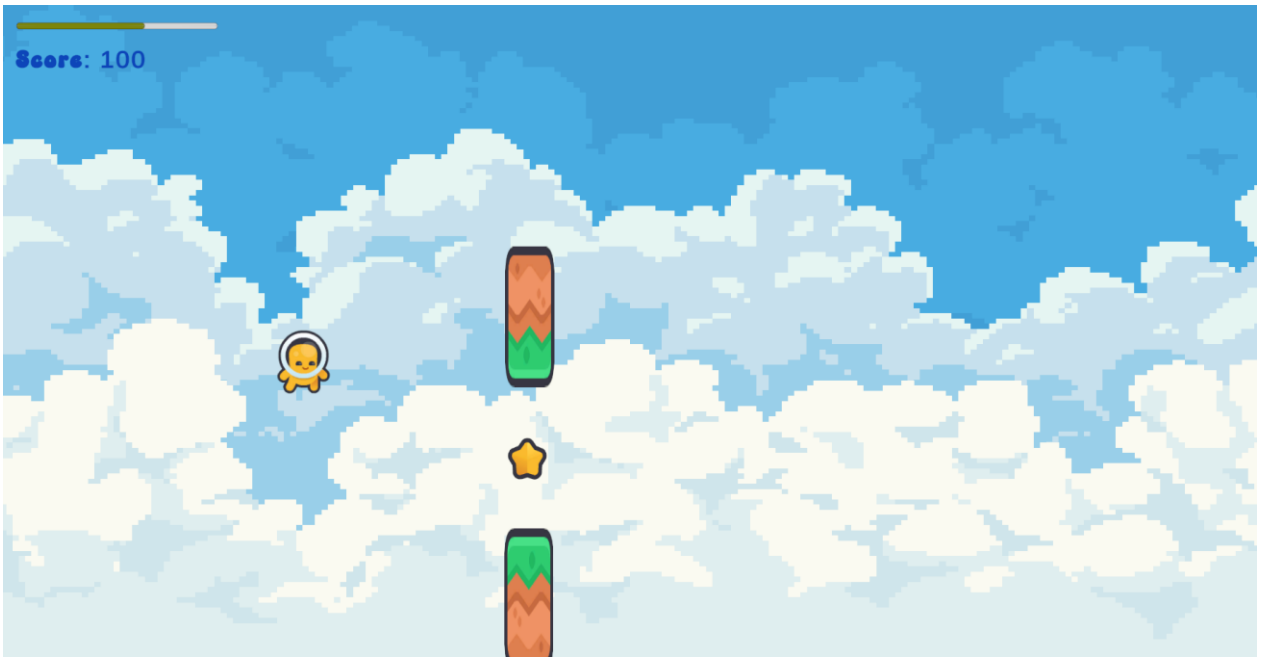
- **Projectile Attacks:** Triggered by a Left Mouse Click, this mechanism allows the player to fire projectiles from their position to defend against incoming threats.

- **Dynamic Enemy AI:** Enemies spawn at varied intervals and heights, forcing the player to make quick decisions to either dodge or destroy them with projectiles.

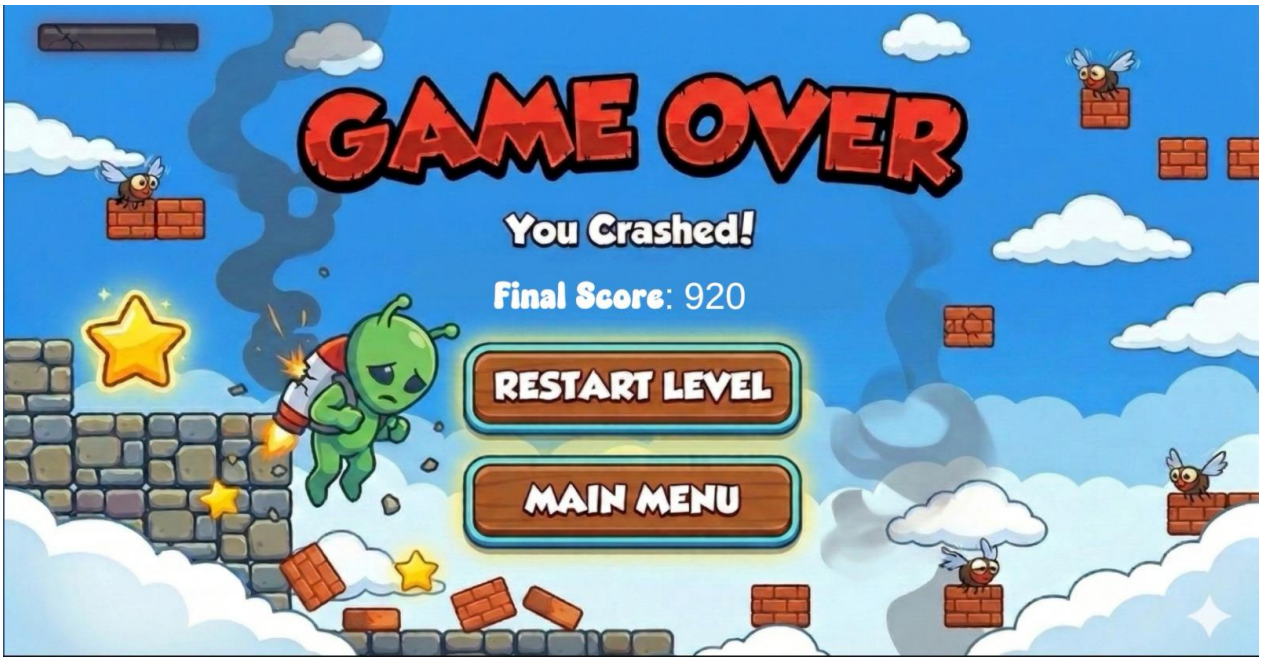
- **Speed Progression:** To ramp up difficulty, the player's forward velocity increases over time, requiring faster reflexes as the game continues.

2. Screenshots of game play

Screenshot 1: Main Menu



Screenshot 2: Gameplay



Screenshot 3: Game Over

3. Control Guide

- **Vertical Movement:** W / S Keys or Up / Down Arrow Keys
- **Shoot Projectiles:** Left Mouse Button (Mouse 0)
- **Restart Game:** Select the "Restart" button on the Game Over screen
- **Exit Game:** Select the "Exit" button on the Main Menu

4. Creative Features

Beyond the core requirements, Star Wings includes the following unique modifications:

- **Weight-Based Spawning:** The ObjectSpawner utilizes a weighted algorithm that evolves. As time passes, enemies and obstacles become more frequent while health packs become rarer, creating a natural difficulty curve.
- **Visual Impact & Feedback:**
  - Screen Shake: Triggered upon collision with obstacles to provide a visceral sense of impact.
  - Dynamic Sprite Rotation: The player sprite changes its tilt (Look Up/Down/Neutral) based on vertical movement to enhance the flight feel.

5. Credits for External Assets/Tools

- **Engine:** Unity Engine
- **UI Rendering:** TextMeshPro
- **Assets:** Sprites and audio clips sourced from the Unity Asset Store and Pixabay

6. Video Link

Gameplay Demo: Click Here